

2 types of DEs: **Differential Equation:** A differential equation is an equation that involves an unknown function $f(x)$ and its derivatives. Common examples include:

1. 1st order DEs:

$$f'(x) = f(x), \quad f''(x) = -f(x).$$

2. Separable Eq.

The order of the differential equation is given by the largest number of derivatives that appear.

↳ Number of derivatives that show up

2. Power series solutions: **Separable Differential Equations:** A separable equation is a first-order differential equation in which

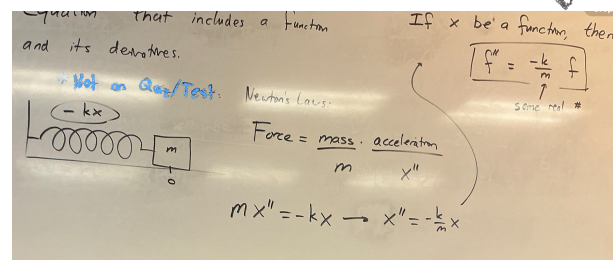
$$\frac{dy}{dx} = g(x)h(y)$$

for $y = f(x)$.

Example: Solve for $y = f(x)$ given

$$y' = \frac{x^2}{y^2} \longrightarrow \frac{dy}{dx} = \frac{x^2}{y^2}.$$

↳ First order bc only 1 derivative



$$\frac{dy}{dx} = \frac{x^2}{y^2} \Rightarrow y^2 \frac{dy}{dx} = x^2 \Rightarrow y^2 dy = x^2 dx \Rightarrow \int y^2 dy = \int x^2 dx$$

$$\frac{1}{3} y^3 = \frac{1}{3} x^3 + C \Rightarrow y^3 = x^3 + 3C \Rightarrow y = \sqrt[3]{x^3 + 3C}$$

bad notationally? To get a unique solution, need a boundary/initial condition. # of boundary conditions for 1st order = order of diff eq. need = order of diff eq.

Ex: $f(\omega) = 2 = \cos^2(30) = \frac{3}{4}$
 $2 = 3 \frac{C}{4} \Rightarrow C = \frac{8}{3}$
 $C = \frac{8}{3} \Rightarrow \frac{1}{3} y^3 = \frac{1}{3} x^3 + \frac{8}{3} \Rightarrow y^3 = x^3 + 8 \Rightarrow y = \sqrt[3]{x^3 + 8}$

So here? we need a boundary condition so we can solve.

you only need 1 C even if you integrate both sides

$\frac{1}{3} C$ could be anything

Note that to get a unique solution, we often have to specify additional information in the form of **boundary conditions**. If $f(0) = 1$, what is the solution to the example above?

Example: Find an expression for $y = f(x)$ with $f(0) = \pi$ given

$$\frac{dy}{dx} = \frac{6x^2}{2y + \cos(y)}$$

$$\frac{dy}{dx} \frac{2y + \cos y}{1} = 6x^2$$

$$2y + \cos y \, dy = 6x^2 \, dx$$

$$\int 2y \, dy + \int \cos y \, dy = \int 6x^2 \, dx$$

$$y^2 + \sin y + C = 2x^3$$

With boundary condition: $y = f(\omega)$, $f(0) = \pi$

$$\pi^2 + \sin(\pi) + C = 0$$

$$C = -\pi^2$$

$$2x^3 = y^2 + \sin(y) - \pi^2 \Leftrightarrow y^2 + \sin(y) = 2x^3 + \pi^2$$

Find expression for $y = f(x)$ w/

$$y' = \frac{x \sec(y)}{y e^{x^2}}$$

$$\frac{dy}{dx} = \frac{x \sec(y)}{y e^{x^2}}$$

$$\frac{dy}{dx} y = \frac{x \sec(y)}{e^{x^2}}$$

$$\frac{dy}{dx} y = \frac{x}{e^{x^2}} \sec(y)$$

$$dy y = \frac{x}{e^{x^2}} dx \sec(y)$$

$$\frac{y}{\sec(y)} dy = \frac{x}{e^{x^2}} dx$$

$$\int \frac{y}{\sec(y)} dy = \int \frac{x}{e^{x^2}} dx$$

$$= \int y \cos(y) dy = \int x e^{-x^2} dx$$

$$u = y \quad dv = \cos y dy$$

$$du = dy \quad v = \sin y$$

$$y \sin y - \int \sin y dy$$

$$y \sin y - (-\cos y)$$

$$= y \sin y + \cos y$$

$$x e^{-x^2} dx$$

$$w = -x^2$$

$$dw = -2x dx$$

$$-\frac{dw}{2} = x dx$$

$$\int -\frac{1}{2} e^w dw$$

$$= -\frac{1}{2} \int e^w dw$$

$$= -\frac{1}{2} e^{-x^2} + C$$

$$y \sin y + \cos y = -\frac{1}{2} e^{-x^2} + C$$

Suppose

$$f(0) = \pi$$

$$\pi \sin(\pi) + \cos(\pi) = -\frac{1}{2} e^0 + C$$

$$-1 = -\frac{1}{2} + C$$

$$\rightarrow \frac{1}{2} = C$$

$$y \sin y + \cos y = -\frac{1}{2} e^{-x^2} - \frac{1}{2}$$

An example of a 2nd order diff eq that's not separable

$$y' = \cos(xy)$$

$$\ln(xy) = \ln(x) + \ln(y)$$

$$\hookrightarrow \frac{dy}{dx} = \ln(x) + \ln(y)$$

Separables rely on multiplication

Still not separable
due to addition

Boundary condition

Example: Solve for $y = f(x)$ with $f(0) = 1$ given

$$y' = \frac{dy}{dx} = x^2 y. \quad \frac{dy}{dx \cdot y} = x^2 \rightarrow \frac{dy}{y} = x^2 dx$$

$$\int \frac{1}{y} dy = \int x^2 dx$$

$$\ln(y) = \frac{x^3}{3} + C \rightarrow e^{\ln(y)} = e^{\left(\frac{x^3}{3} + C\right)} \rightarrow y = e^{\left(\frac{x^3}{3} + C\right)}$$

$$1 = e^{\left(\frac{0^3}{3} + C\right)} = e^C \rightarrow C = 0$$

$$\text{Check: } f'(x) = e^{\frac{x^3}{3}} \cdot \frac{x^3}{3} = x^2 f(x)$$

Find $f(x)$ w/ $f'(x) = f(x)$ and $f(0) = 5$

What's a function whose derivative is itself? e^x

$$f(x) = 5e^x = f'(x)$$

$$\hookrightarrow f(0) = 5e^0 = 5$$

But this isn't very rigorous

power series solutions

Suppose $f(x) = \sum_{n=0}^{\infty} C_n x^n$. Then

$$f'(x) = \sum_{n=1}^{\infty} C_n n x^{n-1}$$

$$f'' = \sum_{n=2}^{\infty} C_n (n)(n-1) x^{n-2}$$

Power series solutions

Suppose $f(x) = \sum_{n=0}^{\infty} C_n x^n$. Then $f'(x) = \sum_{n=1}^{\infty} C_n n x^{n-1}$

$$f'(x) - f(x) = 0 \rightarrow \left(\sum_{n=1}^{\infty} C_n n x^{n-1} \right) - \left(\sum_{n=0}^{\infty} C_n x^n \right) = 0$$

Lemma 1 : If $\sum_{n=0}^{\infty} C_n x^n = 0$, then $C_n = 0$ for all $n = 0, 1, 2, 3$

$$C_n = \frac{f^{(n)}(0)}{n!} = 0 \quad \text{But } C_{n+1} \text{ can be anything}$$

$$\left(\sum_{n=1}^{\infty} C_n n x^{n-1} \right) - \left(\sum_{n=0}^{\infty} C_n x^n \right) = 0$$

$$\hookrightarrow \left(\sum_{n=0}^{\infty} C_{n+1} (n+1) x^n \right) - \left(\sum_{n=0}^{\infty} C_n x^n \right) = 0$$

$$\hookrightarrow \sum_{n=0}^{\infty} (C_{n+1} (n+1) - C_n) x^n = 0 \quad \text{which means that } C_{n+1} (n+1) - C_n = 0 \text{ for all } x$$

So then if $f(0) = 5$ then $C_0 = 5$

$$n=0: C_1 \cdot 1 - C_0 = 0 \rightarrow C_1 = 5$$

$$n=1: C_2 \cdot 2 - C_1 = 0 \rightarrow C_2 = \frac{5}{2}$$

$$n=2: C_3 \cdot 3 - C_2 = 0 \rightarrow C_3 = \frac{5}{6}$$

$$n=1: C_2 \cdot 2 - C_1 = 0 \rightarrow C_2 = \frac{5}{2}$$

$$n=2: C_3 \cdot 3 - C_2 = 0 \rightarrow C_3 = \frac{5}{6}$$

$$\text{So } C_n = \frac{5}{n!}, \text{ then } f(x) = \sum_{n=0}^{\infty} \frac{5}{n!} x^n = 5 \sum_{n=0}^{\infty} \frac{x^n}{n!} = 5e^x$$

A Second-Order Example: Solve for $f(x)$ over the interval $(0, 1)$ with $f(0) = f(\pi) = 0$ given

If $\sum_{n=0}^{\infty} C_n x^n = 0$, then

$C_n = 0$ for all $n = 0, 1, 2, \dots$

$$C_{n+2} = \frac{-\lambda C_n}{(n+1)(n+2)} \text{ for } n \geq 0, 1, 2, \dots$$

$n \geq 0, 1, 2, \dots$

$$C_0, C_2 = \frac{\lambda C_0}{2 \cdot 1}, C_4 = \frac{\lambda^2 C_0}{4 \cdot 3 \cdot 2 \cdot 1}$$

$f''(x) = -\lambda f(x)$. $\lambda = \text{any real number}$

$$\sum_{n=2}^{\infty} C_n (n)(n-1) x^{n-2} + \left(\sum_{n=0}^{\infty} C_n \lambda x^n \right) = 0$$

$$\sum_{n=0}^{\infty} C_{n+2} (n+2)(n+1) x^n + \left(\sum_{n=0}^{\infty} C_n \lambda x^n \right) = 0$$

$$\sum_{n=0}^{\infty} (C_{n+2} (n+2)(n+1) + \lambda C_n) x^n = 0$$

$$C_{2n} = \frac{\lambda^n C_0 (-1)^n}{(2n)!}$$

$$\text{odd } C_1, C_3 = \frac{-\lambda C_1}{3 \cdot 2 \cdot 1}, C_5 = \frac{-\lambda C_3}{5 \cdot 4} = \frac{\lambda^2 C_1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

$$f(x) = \sum_{n=0}^{\infty} C_n x^n = C_0 \sum_{n=0}^{\infty} \frac{(-1)^n \lambda^n x^{2n}}{(2n)!} + C_1 \sum_{n=0}^{\infty} \frac{(-1)^n \lambda^n x^{2n+1}}{(2n+1)!}$$

$$= C_0 \cos(\sqrt{\lambda} x) + C_1 \sin(\sqrt{\lambda} x)$$